16 Lecture 16 (February 21, 2025, est.): Negligible morphisms and semisimplification

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The previous lecture showed that an object on which every Schur functor is nonzero has tensor-power lengths at least $\sqrt{n!}$. This completes the growth argument we will use from the characteristic-zero theory. We now develop a construction that is particularly useful in positive characteristic. Throughout this lecture k is algebraically closed.

16.1 Pseudotensor categories

16.1 Definition. A *symmetric pseudotensor category* over k is an essentially small, additive, k-linear, rigid symmetric monoidal category $\mathcal{C}$ with finite-dimensional morphism spaces and $\mathrm{End}_{\mathcal{C}}(\mathbb{1}) = k$, in which every idempotent splits. The latter condition is also called *Karoubian*: if $e \in \mathrm{End}(X)$ and $e^2 = e$, then $X \cong \mathrm{im}(e) \oplus \mathrm{im}(1-e)$. We do not assume that $\mathcal{C}$ is abelian.

Every additive category has an idempotent completion: its objects are pairs (X, e) with $e^2 = e$, and morphisms $(X, e) \to (Y, f)$ are maps $a : X \to Y$ with $a = fae$. This adds the missing images of idempotents.

16.2 Proposition (Krull–Schmidt)**.** *Every object of $\mathcal{C}$ is a finite direct sum of indecomposable objects, uniquely up to permutation and isomorphism. The endomorphism algebra of an indecomposable object is a finite-dimensional local algebra with residue field k.*

Proof. An orthogonal decomposition of id_X into primitive idempotents in the finite-dimensional algebra $\mathrm{End}(X)$ gives the decomposition of X, because idempotents split. For a primitive idempotent e, the corner algebra $e\,\mathrm{End}(X)e$ is local. Its residue division algebra is finite-dimensional over the algebraically closed field k, hence is k. Uniqueness follows from the usual exchange argument for local endomorphism rings: if id_U is expressed as a sum of composites through the summands of another decomposition, at least one composite is invertible, identifying U with one of those summands; cancel and continue. ∎

In particular, a noninvertible endomorphism of an indecomposable object lies in the radical and is nilpotent. Every endomorphism of such an object is of the form $\lambda\,\mathrm{id} + n$ with $\lambda \in k$ and n nilpotent.

16.2 Tensor ideals and quotient categories

16.3 Definition. A *tensor ideal* $\mathcal{I}$ is a family of linear subspaces $\mathcal{I}(X, Y) \subseteq \mathrm{Hom}_{\mathcal{C}}(X, Y)$ such that

1. if $f \in \mathcal{I}(X, Y)$, $a : W \to X$, and $b : Y \to Z$, then $bfa \in \mathcal{I}(W, Z)$;
2. if $f \in \mathcal{I}(X, Y)$ and $g : U \to V$, then $f \otimes g \in \mathcal{I}(X \otimes U, Y \otimes V)$ and $g \otimes f \in \mathcal{I}(U \otimes X, V \otimes Y)$.

The quotient $\mathcal{C}/\mathcal{I}$ has the same objects and

$$\mathrm{Hom}_{\mathcal{C}/\mathcal{I}}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(X,Y)/\mathcal{I}(X,Y).$$

The ideal conditions make composition and tensor product well defined. The associativity, symmetry, and duality maps descend, giving an additive rigid symmetric monoidal quotient and a quotient functor $Q : \mathcal{C} \to \mathcal{C}/\mathcal{I}$. Every additive symmetric monoidal functor annihilating $\mathcal{I}$ factors uniquely through Q. If an idempotent-complete target is required, one may then take the idempotent completion of the quotient. For the negligible ideal below, the quotient is already semisimple and idempotent complete under the nilpotent-trace hypothesis stated below.

This is a quotient of morphism spaces, not a Serre quotient of an abelian category. In particular, even when both categories happen to be abelian, the quotient functor need not be exact.

16.3 The trace pairing

Recall that for $f : X \to X$, the symmetric categorical trace is

$$\mathrm{Tr}_X(f) = \Big(\mathbb{1} \xrightarrow{\mathrm{coev}_X} X \otimes X^* \xrightarrow{f\otimes \mathrm{id}} X \otimes X^* \xrightarrow{c_{X,X^*}} X^* \otimes X \xrightarrow{\mathrm{ev}_X} \mathbb{1}\Big) \in k.$$

Set $\dim(X) = \mathrm{Tr}_X(\mathrm{id}_X)$. The trace is cyclic, $\mathrm{Tr}_X(gf) = \mathrm{Tr}_Y(fg)$ for $f : X \to Y$, $g : Y \to X$, and is additive on direct sums and multiplicative on tensor products.

16.4 Definition. A morphism $f : X \to Y$ is *negligible* if

$$\mathrm{Tr}_X(gf) = 0 \qquad \text{for every } g : Y \to X.$$

Write $\mathcal{N}(X,Y)$ for the subspace of negligible morphisms. An object is negligible if its identity morphism is negligible.

16.5 Proposition. *The negligible morphisms form a tensor ideal.*

Proof. Linearity is immediate. If $a : W \to X$, $b : Y \to Z$, and $h : Z \to W$, then cyclicity gives $\mathrm{Tr}_W(hbfa) = \mathrm{Tr}_X(ahbf) = 0$, so bfa is negligible. For tensor products, let $f : X \to Y$ be negligible, $g : U \to V$, and $h : Y \otimes V \to X \otimes U$. Close the U-strand of $h(\mathrm{id}_Y \otimes g)$ using evaluation, coevaluation and symmetry; this defines a partial trace $h' : Y \to X$. The trace identities give

$$\mathrm{Tr}_{X\otimes U}\big(h(f \otimes g)\big) = \mathrm{Tr}_X(h'f) = 0.$$

Thus $f \otimes g$ is negligible. Symmetry gives the assertion for $g \otimes f$ as well. ∎

16.4 Benson's lemma

From now on assume additionally that

$$\mathrm{Tr}_X(n) = 0 \quad \text{for every nilpotent } n \in \mathrm{End}_{\mathcal{C}}(X). \tag{16.1}$$

This hypothesis is important for a pseudotensor category. It holds for the representation categories used below, where the categorical trace is the ordinary trace. We use the following criterion, often called Benson's lemma; see also [EO19, §2].

16.6 Lemma. *Let X, Y be nonzero indecomposable objects. A map $f : X \to Y$ is negligible if and only if* $\dim(Y) = 0$ *or f is not an isomorphism. For decompositions $X = \bigoplus_i X_i$ and $Y = \bigoplus_j Y_j$ into indecomposables, a map $f = (f_{ji})$ is negligible if and only if each component $f_{ji} : X_i \to Y_j$ is negligible.*

Proof. Suppose $f : X \to Y$ is not invertible. For any $g : Y \to X$, the endomorphism fg cannot be invertible: otherwise, with $h = (fg)^{-1}$, the map $s = gh : Y \to X$ satisfies $fs = \mathrm{id}_Y$. The nonzero idempotent sf splits X as $Y \oplus Z$. Since X is indecomposable, $Z = 0$, which would make f invertible. Thus fg is a nonunit of the local algebra $\mathrm{End}(Y)$, hence is nilpotent. By (16.1), $\mathrm{Tr}_Y(fg) = 0$.

If f is invertible and $\dim(Y) \neq 0$, taking $g = f^{-1}$ proves that f is not negligible. If $\dim(Y) = 0$, write $fg = \lambda \,\mathrm{id}_Y + n$ with n nilpotent. Then $\mathrm{Tr}_Y(fg) = \lambda \dim(Y) + \mathrm{Tr}_Y(n) = 0$, so f is negligible.

For the last assertion, if all components are negligible, then for $g = (g_{ij}) : Y \to X$,

$$\mathrm{Tr}_Y(fg) = \sum_{i,j} \mathrm{Tr}_{Y_j}(f_{ji} g_{ij}) = 0.$$

Conversely, test against matrices g with just one nonzero component. This isolates each trace pairing and proves that every f_{ji} is negligible. ∎

16.7 Theorem. *The quotient $\overline{\mathcal{C}} = \mathcal{C}/\mathcal{N}$ is a semisimple symmetric tensor category. Its simple objects are the images of the indecomposable objects of $\mathcal{C}$ of nonzero categorical dimension, one for each isomorphism class. An object becomes zero if and only if every indecomposable summand has dimension zero.*

Proof. For nonisomorphic indecomposables X, Y, every map $X \to Y$ is negligible. If $\dim(X) = 0$, the identity of X is negligible, so $Q(X) = 0$. Otherwise Lemma 16.6 identifies the negligible endomorphisms of X with the radical of $\mathrm{End}(X)$, and

$$\mathrm{End}_{\overline{\mathcal{C}}}(Q(X)) = \mathrm{End}_{\mathcal{C}}(X)/\operatorname{rad}\mathrm{End}_{\mathcal{C}}(X) \cong k.$$

Every object is a finite direct sum of these images. Hence the quotient is equivalent, as an additive category, to the category of finite-support families of finite-dimensional vector spaces indexed by the surviving indecomposables. This category is abelian, semisimple and idempotent complete. The quotient retains rigidity and symmetry, and the unit survives because $\dim(\mathbb{1}) = 1$. ∎

16.8 Definition. The category $\overline{\mathcal{C}}$ is the *semisimplification* of $\mathcal{C}$, and Q is its semisimplification functor.

An object of total categorical dimension zero need not become zero: its nonzero-dimensional indecomposable summands may have dimensions whose sum is zero in k. The criterion in the theorem is a condition on every indecomposable summand, not only on their sum.

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